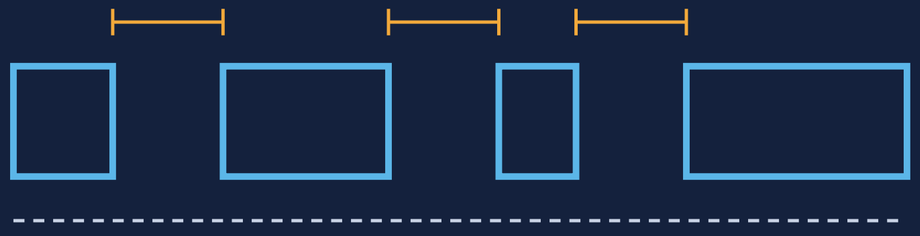
HVAC STUFF · USER GUIDE

Distribute

Elements stand at equal gaps or steps between two borders — in a few clicks



Revit 2023–2027 · Distribute · version 1.3

OVERVIEW

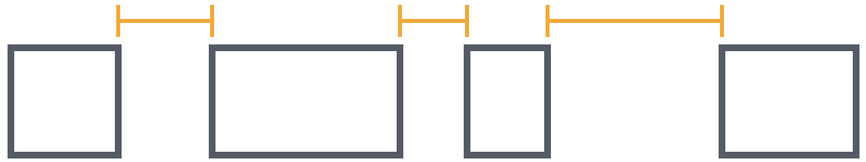
What the commands do

Before



The elements stand as they are: the gaps differ.

After

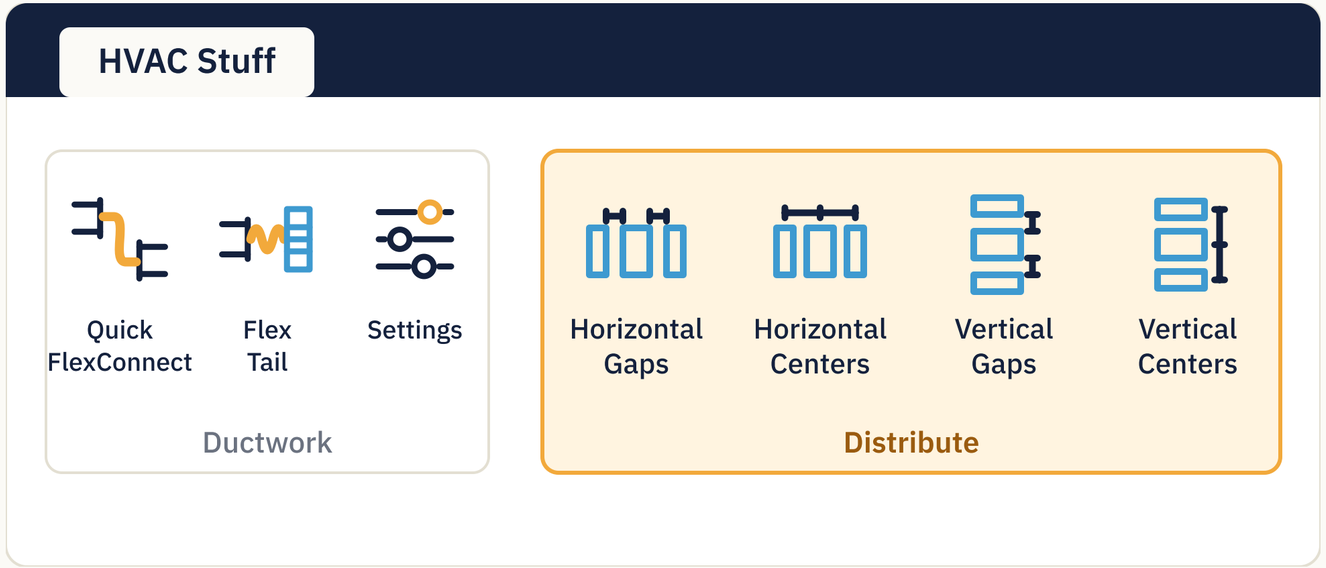


The gaps between edges are equal, the outer ones reach the picked borders.

Select the elements and pick two borders. Ducts, flex ducts, duct fittings, accessories, air terminals and mechanical equipment fit.

WHERE TO FIND IT

Ribbon HVAC Stuff → Distribute



Four buttons on the **Distribute** panel of the HVAC Stuff tab.

They work only in a project, in these views:

- floor, ceiling, or structural plan;
- section and elevation;
- 3D view without perspective.

In families, on sheets and in perspective the buttons are gray.

QUICK START

Three steps to an even layout

1

Select the elements

Select at least two elements and click the button. Or click the button first, then select them and click Finish.

2

Pick two borders

Hover a face, a drawing line, a wall or a grid: the nearest one is marked; click takes it. Twice.

3

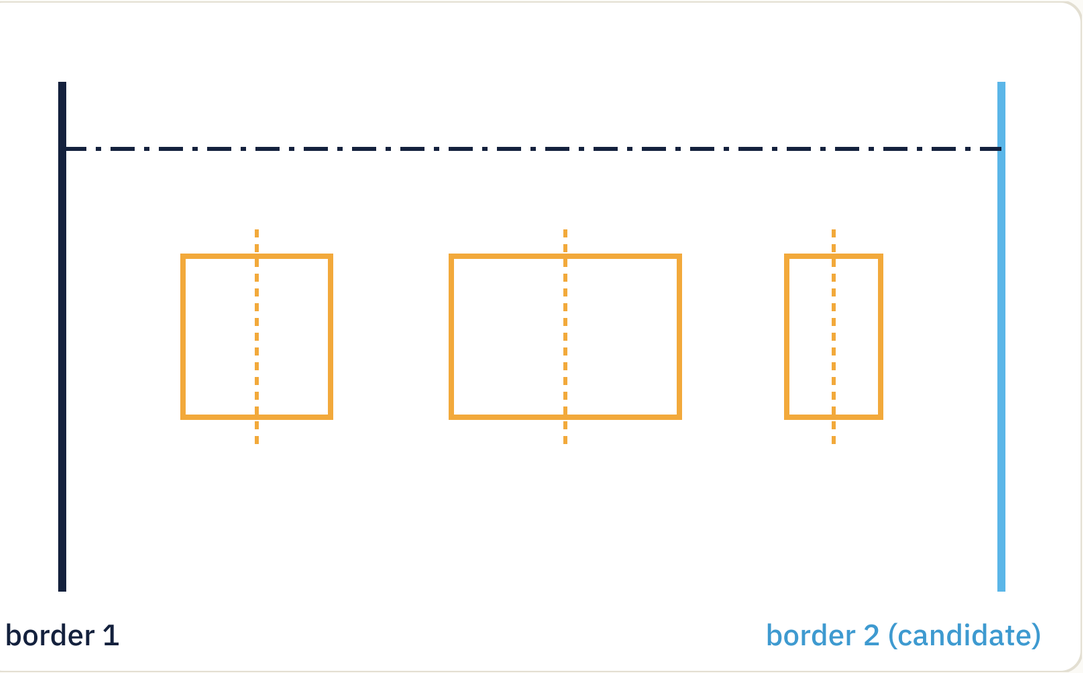
Done

The elements move in one undo step. An error or a warning goes to the message window and nothing moves.

Esc at any step cancels. The elements move along the horizontal (or vertical) direction of the active view; the points only give the start and the end.

TWO BORDERS

Borders: hover, like connectors



Hover, no need to aim. The face, imported drawing line, linked-model wall or grid nearest to the cursor is marked; a click takes it.

A dash-dot line follows the cursor from the first border and shows the distance.

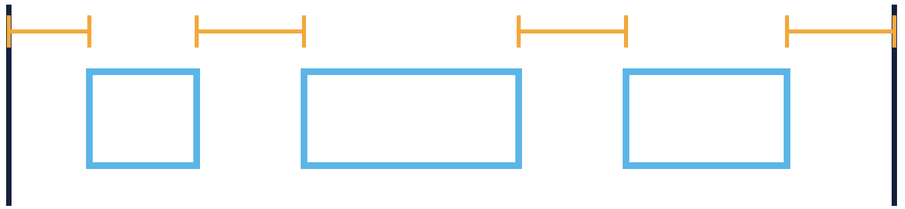
Colors: orange — the edges and the center of every element; dark blue — the picked borders; light blue — the candidate under the cursor. The marks disappear when the command ends.

Free points: turn on “Pick borders as free points” in Settings → Distribute: the borders are then picked as any point instead of snapping (a PDF underlay has no geometry).

FOUR COMMANDS

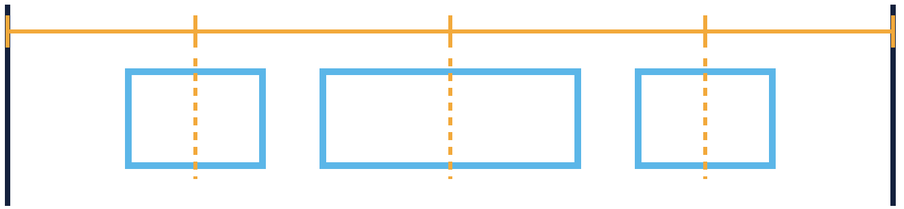
Gaps or centers — horizontally or vertically

Gaps



Element sizes count: the sizes of all elements are subtracted from the distance and the rest is divided by the number of elements plus one. The same gap lies between a point and an element and between neighbors.

Centers



Sizes are ignored: the centers stand at equal steps, the same step to each point.

The Horizontal commands move elements along the horizontal of the view, the Vertical ones along the vertical. The points need not lie on one line: only their projection on that direction counts.

KEYS

Shift and Ctrl — while you click

Shift dimensions

Also places dimensions: one per interval, including those between the borders and the nearest elements. Not in 3D.

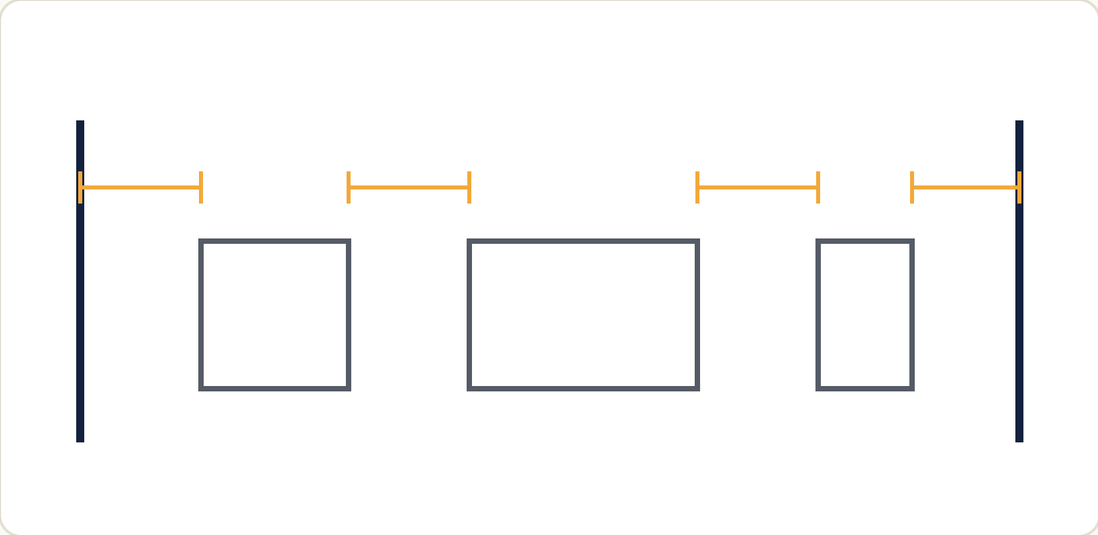
Ctrl align

Also aligns the elements across the direction to the first one, so they stand in one line.

Hold the key while you click the ribbon button or while you pick the borders. The keys can be combined.

SHIFT

Dimensions between the elements



One dimension per interval, including the intervals between the picked borders and the nearest elements.

The end dimensions attach to what is found near the point: a face in the model, a wall of a linked model, a drawing line; with snapping, the picked borders themselves.

No face or edge — the dimension runs from the element axes; the message window says so.

Nothing to dimension from (a drawing polyline, a PDF underlay, a free point): a tiny helper detail line (1 mm on the sheet) is created at the border and the dimension runs from it. The message says how many lines were created; delete them if you do not need them.

TRICKY CASES

Joined elements, ducts and undo

Joined elements move as one

Elements connected directly to each other (a static pressure box with its diffuser) move as one unit. The size and the center of the unit are those of its air terminals, so the diffuser stands where the calculation puts it.

Ducts between them

Revit stretches the ducts between the elements itself. The command checks the result and lists the elements that did not end up at their calculated place, with the largest deviation in millimeters.

One undo step

Everything moves in one transaction. Pinned elements, elements without geometry and Revit failures are reported in the message window, and nothing moves.

MESSAGES

What the message window shows

Window title	What it says
Select at least two elements	Ducts, duct fittings, duct accessories, air terminals and mechanical equipment can be distributed. Suitable elements selected: N.
The two points coincide	The points define the axis along which the elements move. Pick two different points.
The elements do not fit between the points	The elements are N mm wider in total than the distance between the points (M mm). Move the second point farther away, select fewer elements, or use a Centers command.
Pinned elements cannot be moved	Unpin the elements and run the command again. Nothing was moved.
Cannot measure some elements	These elements have no geometry that can be measured. Nothing was moved.
The selected elements are joined into one group	Elements connected directly to each other move together, so there is nothing to spread. Select elements that are not connected to each other (ducts between them are fine).
Some elements are not at their calculated positions	Elements connected to the selected ones moved them as well, or Revit limited the move. Largest deviation: N mm.

IF SOMETHING GOES WRONG

Frequently asked questions

Situation	What to do
The buttons are gray	Open a plan, section, elevation or a 3D view without perspective in a project.
An element is not taken into the selection	Only ducts, fittings, accessories, air terminals and equipment fit; the rest is skipped and the message says how many.
A border does not snap	A PDF underlay or a curve without geometry: turn on free points in Settings → Distribute and pick them.
“The elements do not fit”	The sizes add up to more than the distance: move the points apart, select fewer elements or use a Centers command.
No dimensions	Hold Shift. Dimensions are not placed in a 3D view — use a plan, a section or an elevation.
An error	Click Copy in the message window and send the text to the developer.

HVAC STUFF

Two borders — and everything stands even

New versions, installers and guides: github.com/iMrVoLaNd/HVAC-Stuff-Releases

Help from Revit: point at the button and press **F1**.

Also in the add-in: Quick FlexConnect and Flex Tail.